

DEPARTMENT OF MATHEMATICS & STATISTICS
MTH 211: THEORY OF STATISTICS (2025-26)
Quiz 1 – Solutions

Problem 1

The likelihood ratio or sufficiency argument:

$$\frac{f_{\theta}(\underline{x})}{f_{\theta}(\underline{y})} = \frac{\prod_{i=1}^n I_{(\theta, \theta+1)}(x_i)}{\prod_{i=1}^n I_{(\theta, \theta+1)}(y_i)} = \frac{I_{(\theta, \infty)}(x_{(1)}) I_{(-\infty, \theta+1)}(x_{(n)})}{I_{(\theta, \infty)}(y_{(1)}) I_{(-\infty, \theta+1)}(y_{(n)})}$$

If $x_{(1)} = y_{(1)}$ and $x_{(n)} = y_{(n)}$, then the ratio is 1, which is independent of θ for all $\theta \in \mathbb{R}$. Hence, $T(\underline{X}) = (X_{(1)}, X_{(n)})$ is a **Minimal Sufficient Statistic (MSS)**.

Ancillary Statistic: Note $Y_i = X_i - \theta$ for $i = 1, \dots, n$ are i.i.d. $U(0, 1)$. Then the range $X_{(n)} - X_{(1)} = (X_{(n)} - \theta) - (X_{(1)} - \theta) = Y_{(n)} - Y_{(1)}$ has a distribution that does not depend on θ . Thus, $X_{(n)} - X_{(1)}$ is an ancillary statistic.

Completeness Check: Hence, we know that for $U(\theta, \theta + 1)$:

$$E_{\theta}[X_{(n)} - X_{(1)}] = a + b_n \quad \left(\text{here, in particular, } a = 0, b_n = \frac{n-1}{n+1}, \text{ but no need to derive explicitly} \right)$$

Let $h(x_{(1)}, x_{(n)}) = x_{(n)} - x_{(1)} - (a + b_n)$. Then $E_{\theta}[h(X_{(1)}, X_{(n)})] = 0$ for all θ , but $h(x_{(1)}, x_{(n)}) \neq 0$ almost everywhere. Hence, the MSS $(X_{(1)}, X_{(n)})$ is **not complete**.

Common Mistakes :

1. Not able to identify the MSS correctly, mostly settling for $X_{(n)}$
2. Those who do not know the location shift argument, making mistakes in derivation of $E_{\theta}[X_{(n)} - X_{(1)}]$
3. Wrong understanding of the definition of completeness.

Problem 2

Given $f_{\theta}(x) = \frac{1}{\sqrt{2\pi\theta}} \exp\left(-\frac{(x-\theta)^2}{2\theta}\right)$, this is an exponential family distribution:

$$f_{\theta}(x) = \frac{1}{\sqrt{2\pi\theta}} \exp\left(-\frac{1}{2\theta} \sum x_i^2 + \sum x_i - \frac{n\theta}{2}\right)$$

Parameters:

- $\eta(\theta) = \eta = -\frac{1}{2\theta}$
- $T(x) = \sum x_i^2$
- $B(\theta) = \frac{1}{2}(\theta + \ln \theta)$

The natural parameter space $H = \{\eta : \eta < 0\}$ **contains an open rectangle** (open interval) in $\mathbb{R}^1$, indicating a full rank exponential family. Hence, the MSS $T(X) = \sum X_i^2$ is **complete**.

Finding the UMVUE for θ^2 :

Multiple correct answers are possible; one such approach is given below.

$$E\left(\sum X_i^2\right) = \sum E(X_i^2) = \sum [V(X_i) + E^2(X_i)] = n(\theta + \theta^2)$$

$$E\left(\sum X_i^2 - \sum X_i\right) = n\theta + n\theta^2 - n\theta = n\theta^2$$

Therefore, $\frac{1}{n} \sum_{i=1}^n (X_i^2 - X_i)$ is the **UMVUE** of θ^2 .

Common Mistakes:

Many people have written the MSS as $T(X) = (\sum X_i, \sum X_i^2)$. This is incorrect; the $\sum X_i$ term does not have any function of θ associated with it. The same problem was a part of a problem set. It was discussed in class and solution was uploaded in Mookit!

Some have given trivial unbiased estimator based on a single observation. It was clearly asked to give an estimator based on all observations.

Problem 3

Given a Geometric Distribution: $P(X = x) = (1 - p)^{x-1}p$ for $x = 1, 2, \dots$. To find the UMVUE of $g(p) = p$:

$$f_p(x) = \exp((x-1)\ln(1-p) + \ln p)$$

This is an exponential family with $\eta_1 = \ln(1-p)$ and $T(X) = \sum X_i$. The natural parameter space $H = \{\eta : \eta < 0\}$ **contains an open rectangle** (open interval), ensuring completeness. The MSS $T = \sum X_i$ is complete.

Common Mistakes(Part A):

Students have not mentioned the existence of an **open rectangle** for the full rank exponential family justification.

Let $\delta(\underline{x}) = 1$ if $x_1 = 1$, and 0 otherwise. Then $E[\delta(\underline{x})] = P(x_1 = 1) = p$. By Lehmann-Scheffé theorem, the UMVUE is $E[\delta(\underline{x})|T = t]$:

$$\begin{aligned} P(x_1 = 1 | \sum x_i = t) &= \frac{P(x_1 = 1, \sum_{i=2}^n x_i = t-1)}{P(\sum_{i=1}^n x_i = t)} \\ &= \frac{p \cdot \binom{(t-1)-1}{(n-1)-1} p^{n-1} (1-p)^{(t-1)-(n-1)}}{\binom{t-1}{n-1} p^n (1-p)^{t-n}} = \frac{\binom{t-2}{n-2}}{\binom{t-1}{n-1}} = \frac{n-1}{t-1} \end{aligned}$$

The unique UMVUE is $\hat{p} = \frac{n-1}{T-1}$ for $T \geq n, n > 1$.

Common Mistakes(Part B):

Everybody has left this part without writing the final simplified UMVUE expression - couldn't realise that $T = \sum_{i=1}^n X_i$ is total number of trials required to get n successes and hence follows Neg Bin(n, p) with pmf $P(T = t) = \binom{n-1}{t-1} p^n (1-p)^{t-n}$, $t = n, n+1, n+2, \dots$